

Name: _____

Lab Exercise: Spreading Bacteria using Glass Beads

Plating beads are designed to spread 50-400 μl of bacterial or yeast cells onto an agar plate. They provide an alternative to the traditional method of using a spreader or “hockey stick.”

Purpose of the Exercise

You will utilize sterile glass beads to uniformly distribute bacteria on solid media.

Learning Objectives - After completing the exercise, you will be able to:

1. Utilize glass beads to spread bacteria.
2. Identify the purpose of utilizing glass beads.

Procedure:

1. Gather materials
 - a. Agar plate
 - b. Sterile glass beads
 - c. Sample to transfer
2. Label the agar plate appropriately.
3. Aseptically inoculate plate with specified volume of sample (50-400 μl) by lifting the lid from a plate just enough to allow you to position the tip of the pipet over the center of the agar surface and deliver the sample. Replace the lid immediately.
4. Open the foil cover and pour 5-10 sterilized glass beads onto the agar surface of the plate. Quickly replace the lid.
5. Keeping the bottom of the plate in contact with the lab bench or counter, vigorously shake the plate in at least two directions for 10 to 15 seconds to move the beads randomly over the surface.
6. Pick up the plate, turn it over with bottom up, and tap the bottom so that the beads drop onto the inside of the lid.
7. Over a beaker or container containing a bleach solution, remove the lid and turn both the lid and the plate bottom so that the inside surfaces face down (minimizing the risk that a contaminant might land on the surface).
8. Drop the beads into the beaker. Quickly replace the lid. Do not discard the beads. They are re-sterilized and re-used.
9. Invert and incubate as directed.

